Machine Learning Report

Dataset

The dataset consists of 569 samples which get their description from 30 numerical predictors that analyze tumour characteristics through digitized cell nuclei images. The target variable diagnosis received binary encoding through "B" (benign) and "M" (malignant) labels which became 0 and 1 respectively to create a binary classification problem (Han et al., 2012). The data analysis showed no missing data points but the "worst" measurements followed a right-skewed distribution and size-related features displayed substantial multicollinearity. Ensemble and neural approaches function independently from multicollinearity according to Hastie et al. (2009). The dataset maintains equal numbers of benign and malignant cases at 63% versus 37% so I applied an 80/20 train-test split with stratification to preserve class balance. The medical classification dataset contains useful data but requires special treatment for its typical problems with outliers and unbalanced data and correlated features.

Algorithms Used

The research study employed three supervised learning methods for analysis. The interpretable baseline model employed Logistic Regression to calculate malignancy odds through linear combinations of features. The research implemented two types of regularization to prevent overfitting through L2 regularization with C=1.0 and C=0.1 and L1 regularization using the liblinear solver (Hosmer et al., 2013).

Random Forest served as the second traditional model which Breiman (2001) developed. The decision tree training process on bootstrapped samples generates multiple trees which produce combined predictions to achieve robustness against multicollinear data and non-linear relationships. The evaluation of three configurations involved changing tree numbers from 100 to 300 and setting maximum tree depths at None, 8 and 5.

The third model used a Neural Network (MLPClassifier) which employed multilayer perceptrons for backpropagation-based training (Goodfellow, Bengio & Courville, 2016). The research investigated three different neural network architectures which included a single hidden layer with 16 ReLU neurons and two hidden layers with 32 and 16 ReLU neurons and a Tanh-based network with enhanced regularization strength (alpha=0.001). The StandardScaler preprocessing method was used to handle neural network sensitivity to feature scale (Pedregosa et al., 2011).

Results

All models performed strongly. The F1-score of Logistic Regression reached 0.9756 when L2 regularisation was applied with C=0.1. Random Forest models achieved F1-scores between 0.95 and 0.96 which demonstrated their ability to detect non-linear patterns but showed lower recall accuracy. The single-hidden-layer neural network achieved the best results among all models with accuracy = 0.9825 and precision = 1.0 and recall = 0.9524 and F1 = 0.9756 and ROC-AUC = 0.9970. The research of Shavitt and Segal (2018) confirms our results because simple neural networks outperform complex models when processing structured tabular data.

Suggestions for Improvement

The model performance will increase when PCA or feature selection techniques remove multicollinearity and decrease data noise. The model will detect malignant cases more effectively through the use of SMOTE or ADASYN oversampling techniques which address class imbalance problems (Chawla et al., 2002). The models will reach their best performance through extensive hyperparameter optimization conducted by grid or random search methods. The model will achieve better stability through two methods:

log transformation of skewed predictors and stacked generalization which combines linear and ensemble and neural models for enhanced performance.

References

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